

WHAT IS PALEONTOLOGY?

Paleontology is the study of prehistoric life. It combines several fields, including geology (study of rocks), biology (study of living things), and chemistry (study of the basic building blocks of matter), to help piece together the evidence collected from fossils and the surrounding rocks. This evidence is usually incomplete because many important details are not preserved during the fossilization process. To make up for these gaps in information, paleontologists study fossils of similar species, as well as modern-day animals, to help them build a picture of prehistoric creatures and understand how they might have lived.

Paleontologists use small brushes and chisels to carefully uncover the fossils.

INTERPRETING FOSSILS

Scientists are frequently uncovering new information that changes their understanding of prehistoric animals. For example, for a long time, a spiky bone found with the fossils of the herbivore *Iguanodon* was thought to be part of the dinosaur's nose. However, research later revealed that the spike was actually part of the creature's thumb. Paleontologists are constantly having to think again about their understanding of prehistoric creatures when new evidence such as this comes to light.



BURIED BONES

Digging for fossils can be hard work. Teams of excavators sometimes have to endure tough conditions such as searing heat or freezing wind. This fossil hunt conducted by the Great Plains Dinosaur Museum and Field Station, Montana, USA, shows experts carefully chipping away at the rocks to reveal the bones of a large dinosaur.

The sediments brushed off from the bones are collected to check for small bits of fossils.

